

# BOOKS ANALYSIS CAPSTONE PROJECT

Web Scraping • Data Cleaning • Exploratory Analysis • K-Means Clustering • Power BI

Final Report / Presentation Document by Y.Nikhil

**Project focus: transform publicly available book data into structured insights about price, rating, availability and book segments, then communicate the model outcomes through an interactive Power BI dashboard.**

Prepared for submission against the Data Science PGC Capstone Project brief.

## 1. Executive Summary

This project presents an end-to-end data science workflow using publicly available book information collected from the web. The workflow covers web data acquisition, cleaning and feature engineering, exploratory data analysis, K-Means clustering, and interactive reporting in Power BI. The project was designed to turn raw scraped records into decisionready information about book pricing, ratings and meaningful book segments.

The working dataset contains 1,000 book records with fields including title, price, rating, availability and URL. Additional analytical fields were created for cleaned price, numeric rating, price category, rating category and cluster name. The final clustering solution uses four segments: Budget Highly-Rated, Budget Low-Rated, Premium HighlyRated and Premium Low-Rated.

The Power BI report is structured around an Overview dashboard, a Model Analysis page and a Book Details drillthrough page. The dashboard includes KPI cards, price and rating distributions, a price-versus-rating cluster visualization, slicers and a clear-all-filters control. The planned Model Analysis page adds K-Means visuals, Key Influencers as an AI visual, and a Word Cloud as a custom visual.

**The project brief requires an end-to-end workflow, a minimum of 1,000 clean records, model development, an interactive dashboard with KPIs, AI/custom visuals and reporting features, and a final PDF report summarizing methodology, insights, model results and recommendations. This report is organized to cover those deliverables.**

## 2. Business Problem & Objectives

### Business Problem

Online book catalogues contain useful information, but raw web data is usually unstructured and difficult to compare directly. The business problem is to transform scraped book-level information into an analytical dataset that can reveal pricing patterns, rating patterns and groups of books with similar characteristics.

### Objectives

- Collect at least 1,000 book records from a publicly accessible web source.
- Clean inconsistent price and rating values and prepare analysis-ready features.
- Explore price, rating and category distributions through visual analysis.
- Apply K-Means clustering to segment books using price and rating characteristics.
- Interpret cluster profiles using average price, average rating and cluster size.
- Build an interactive Power BI dashboard that communicates key insights and model outcomes.
- Demonstrate reporting features such as drill-through, bookmarks, synchronized slicers and RLS.

### Success Criteria

- A clean 1,000-record dataset suitable for analysis.
- A reproducible scraping and preprocessing/modeling workflow.
- A defensible clustering solution with interpretable segments.
- A Power BI report that can be used interactively by a reviewer.
- A final report and presentation that clearly communicate methodology, findings and recommendations.

## 3. Data Acquisition & Dataset

The project follows the capstone requirement to use publicly accessible web data and collect at least 1,000 clean records. The working book dataset contains 1,000 entries and the core scraped fields are Title, Price, Rating, Availability and URL.

| Field           | Purpose / use                                                             |
|-----------------|---------------------------------------------------------------------------|
| title           | Book title; used for book-level identification and the custom Word Cloud. |
| price           | Original scraped price text.                                              |
| price_clean     | Numeric price used for analysis and clustering.                           |
| rating          | Original rating label.                                                    |
| rating_numeric  | Numeric rating used for analysis and clustering.                          |
| availability    | Availability information from the source.                                 |
| url             | Source link for book-level drill-through details.                         |
| price_category  | Derived price grouping: Low / Medium / High.                              |
| rating_category | Derived rating grouping: Low Rated / Average / Highly Rated.              |
| cluster         | K-Means cluster identifier.                                               |

|              |                                         |
|--------------|-----------------------------------------|
| cluster_name | Business-friendly K-Means segment name. |
|--------------|-----------------------------------------|

The project brief explicitly asks for multiple useful fields per record and a clean, consistent dataset ready for analysis. The fields above support both the analytical model and the Power BI reporting layer.

## 4. Data Cleaning & Feature Engineering

### Key preprocessing activities

- Standardized the scraped price field by removing the pound symbol and converting the value to numeric format.
- Converted rating words into numeric values so that ratings could be compared and used in modeling.
- Created price\_clean as a numeric analysis field.
- Created rating\_numeric as a numeric rating field.
- Created price\_category to group books into Low, Medium and High price bands.
- Created rating\_category to group books into Low Rated, Average and Highly Rated.
- Created cluster and cluster\_name fields after model development.
- Checked for duplicate titles; one duplicate title was identified in the working dataset.

**These transformations convert web-scraped text into structured variables that can be aggregated, visualized and supplied to a machine-learning model.**

## 5. Exploratory Data Analysis

### Price distribution

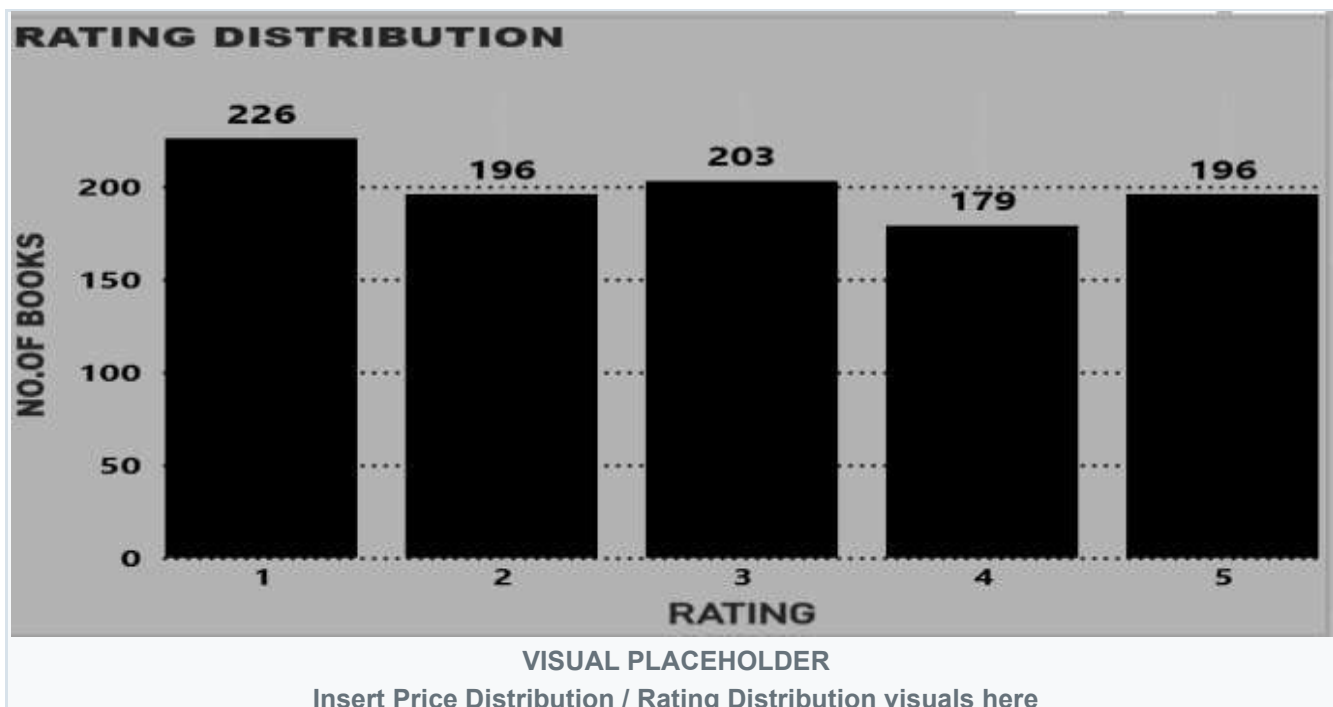
The price-category distribution contains 403 High-price books (40.3%), 401 Medium-price books (40.1%) and 196 Low-price books (19.6%). The distribution indicates that the dataset is concentrated in the Medium and High price categories.

### Rating distribution

The numeric rating counts are: 1-star = 226, 2-star = 196, 3-star = 203, 4-star = 179 and 5-star = 196. The largest group is rating 1, while rating 4 has the smallest count among the five rating levels.

### Price and rating relationship

The Power BI scatter plot places price\_clean on the x-axis and rating\_numeric on the y-axis, with cluster\_name used as the legend. This visual connects the exploratory analysis directly to the K-Means model outcomes.



## 6. Model Building & Evaluation

K-Means clustering was selected to segment books based on quantitative characteristics. The main modeling variables used in the dashboard analysis are price\_clean and rating\_numeric. The model assigns each book to a cluster, after which cluster names are interpreted using the average price and average rating profiles.

### Cluster evaluation

Silhouette-score checks were performed for different K values. The recorded results included a score of approximately 0.429 for K=4 and approximately 0.436 for K=8. The final reporting solution uses four business-interpretable clusters,

| Cluster              | Book count | Average price | Average rating | Business interpretation                        |
|----------------------|------------|---------------|----------------|------------------------------------------------|
| Budget Highly-Rated  | 308        | £21.46        | 3.92           | Lower-priced books with relatively strong r    |
| Budget Low-Rated     | 197        | £21.97        | 1.45           | Lower-priced books with relatively low rating  |
| Premium Highly-Rated | 270        | £47.54        | 4.05           | Higher-priced books with strong ratings        |
| Premium Low-Rated    | 225        | £45.81        | 1.48           | Higher-priced books with relatively low rating |

allowing the dashboard to communicate the segmentation clearly.

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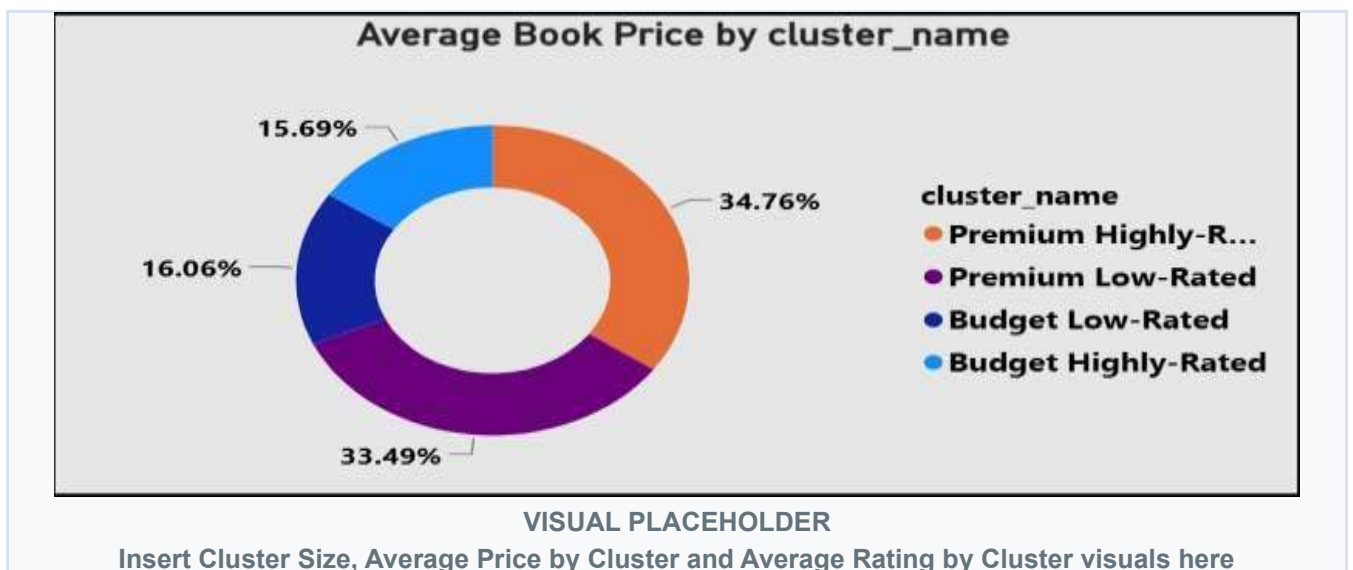
The cluster profiles show a clear separation between lower-price and higher-price books, while rating quality creates a second dimension. This makes the four-cluster solution easy to explain to a non-technical reviewer.



## 7. Model Insights & Interpretation

- Premium Highly-Rated is the strongest cluster on both average price and average rating, making it a natural example of premium books with strong perceived quality.
- Budget Highly-Rated combines a lower average price with a relatively strong average rating, which can be viewed as a value-oriented segment.
- Budget Low-Rated and Premium Low-Rated both have average ratings close to 1.5, indicating that rating level separates these segments from the highly-rated groups.
- The model therefore provides a two-dimensional segmentation: price positioning and rating performance.
- The cluster labels are business-friendly interpretations of the numeric K-Means assignments and should be treated as analytical segment names rather than source-provided categories.

### Recommended visual evidence



## 8. Interactive Power BI Dashboard

The Overview page communicates the main book-level insights through four KPI cards and multiple analytical visuals. The current design includes Total Books (1K), Average Book Price (£35.07), Average Rating (2.92) and Number of Clusters (4).

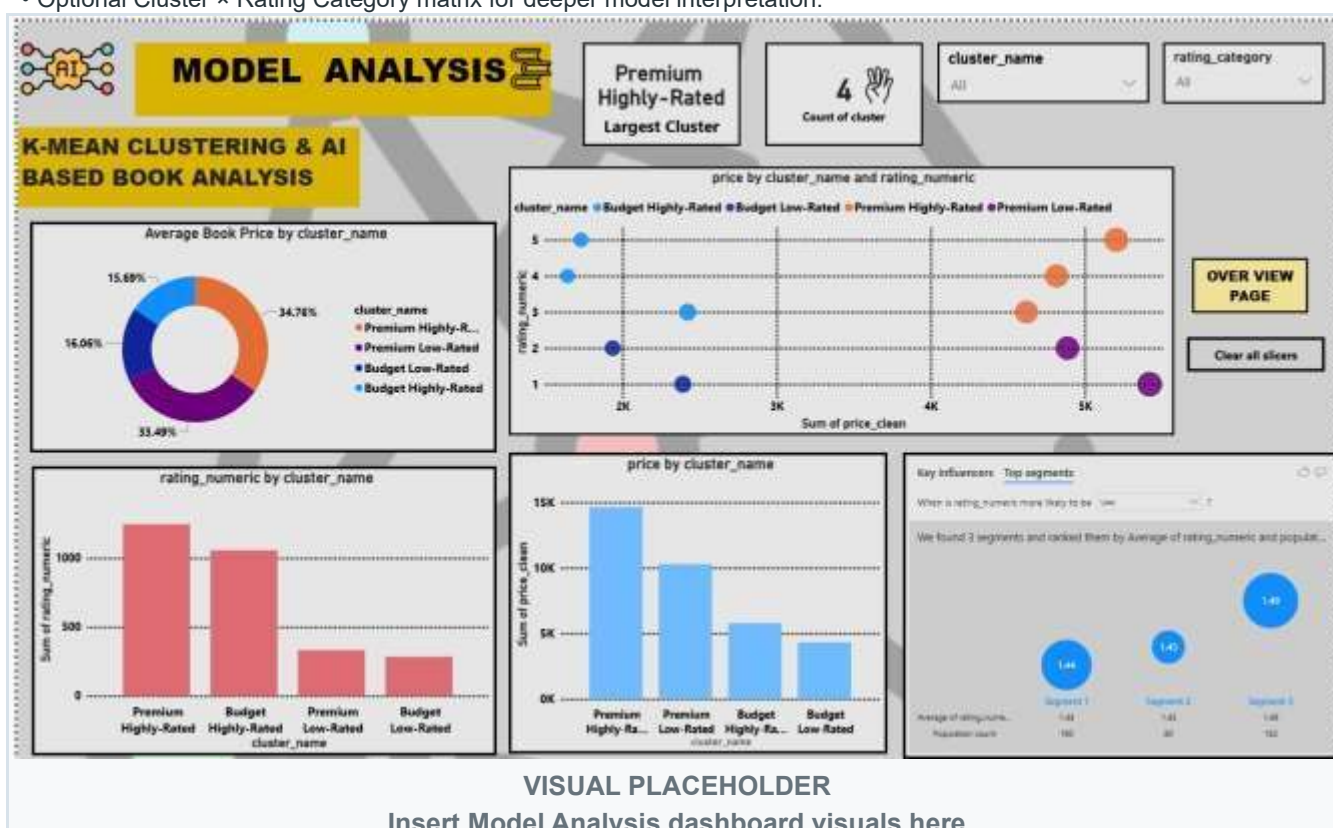
| Dashboard element      | Purpose                                                   |
|------------------------|-----------------------------------------------------------|
| Total Books KPI        | Shows dataset size.                                       |
| Average Book Price KPI | Summarizes the overall average cleaned price.             |
| Average Rating KPI     | Summarizes the overall numeric rating.                    |
| Clusters KPI           | Shows the number of model segments.                       |
| Price Distribution     | Shows the Low / Medium / High price mix.                  |
| Rating Distribution    | Shows counts across ratings 1–5.                          |
| Price vs Rating        | Shows the relationship between price, rating and cluster. |
| Cluster slicer         | Filters the dashboard by model segment.                   |
| Rating Category slicer | Filters by rating grouping.                               |
| Price Category slicer  | Filters by price grouping.                                |
| Clear all slicers      | Provides a quick filter reset.                            |



## 9. Model Analysis Board — Planned Visuals

The Model Analysis page is designed to make the machine-learning component visible to the evaluator rather than hiding it behind the main dashboard. It combines model KPIs, cluster comparisons, the K-Means scatter plot, an AI visual and a custom visual.

- KPI: Number of Clusters.
- KPI: Largest Cluster.
- KPI: Highest Rated Cluster.
- K-Means Book Segmentation scatter chart using price\_clean, rating\_numeric and cluster\_name.
- Books by Cluster bar chart using cluster\_name and count of title.
- Average Price by Cluster column chart.
- Average Rating by Cluster column chart.
- Key Influencers AI visual to explore factors associated with rating\_numeric.
- Word Cloud custom visual using title.
- Optional Cluster × Rating Category matrix for deeper model interpretation.



## 10. Power BI Reporting Features Checklist

The capstone brief specifically asks for reporting features including RLS, synchronized slicers, bookmarks and drillthrough. The following implementation plan should be completed before final submission.

| Feature       | Implementation in this project                                                      | Status      |
|---------------|-------------------------------------------------------------------------------------|-------------|
| RLS           | Create demonstration roles using cluster_name to restrict visible rows.             | To complete |
| Sync slicer   | Synchronize Cluster, Rating Category and Price Category slicers across report page. | To complete |
| Bookmark      | Create Overview and Model Analysis navigation states/buttons.                       | To complete |
| Drill-through | Create Book Details page with title as the drill-through field.                     | To complete |

|               |                                     |         |
|---------------|-------------------------------------|---------|
| AI visual     | Key Influencers on rating_numeric.  | Working |
| Custom visual | Word Cloud using title.             | Planned |
| KPIs          | Four overview KPIs plus model KPIs. | Working |

**Note on trend charts: the working book dataset does not contain a meaningful date/time dimension. A fabricated time trend should therefore be avoided. The report instead emphasizes distributions, relationships and model segmentation. If a valid time field is later added to the dataset, a true trend chart can be included.**

## 11. Business Insights & Recommendations

### Insights

- The dataset is weighted toward Medium and High price books, with more than four-fifths of records falling into those categories.
- Ratings are spread across all five levels, with rating 1 having the highest count in the working dataset.
- The clustering model identifies four interpretable book segments based on price and rating.
- Premium Highly-Rated books have the highest average price and highest average rating among the four segments.
- Budget Highly-Rated books indicate a potentially attractive value segment because the average rating is strong while the average price is comparatively low.

### Recommendations

- Use the Premium Highly-Rated segment as a premium recommendation category when quality and price are both important.
- Use Budget Highly-Rated as a value-for-money category for users who prioritize strong ratings at lower prices.
- Use the low-rated segments as candidates for additional quality review rather than treating price alone as a proxy for quality.
- Keep cluster-level filtering available in the dashboard so stakeholders can compare segments interactively.
- Use the Book Details drill-through to move from high-level patterns to individual books and source URLs.
- Extend the dataset in future iterations with additional dimensions such as genre, author, publication date or sales-related measures if legitimately available.

## 12. Limitations & Future Scope

- The dataset represents a scraped catalogue snapshot rather than historical sales performance.
- The absence of a valid time dimension limits true trend analysis.
- K-Means clustering is based primarily on price and rating, so other book characteristics are not represented in the current segmentation.
- The cluster names are interpretation labels created for reporting and are not source categories.
- The dashboard should be refreshed when the underlying scraped dataset changes so that model and KPI outputs remain consistent.

### Future enhancements

- Add a valid date dimension and create time-based trend analysis if future scraping captures dates.
- Evaluate clustering with additional standardized numeric features when appropriate.
- Add genre or author-level analysis if those fields are available in the source.
- Create a richer recommendation layer using cluster membership and rating/price preferences.
- Add more Power BI navigation and security controls for a production-style reporting experience.

## 13. Final Submission Checklist

| Deliverable                | Prepared / required                                             |
|----------------------------|-----------------------------------------------------------------|
| Web scraping script        | Include the reproducible Python extraction script.              |
| Cleaning + modeling script | Include preprocessing, feature engineering and K-Means code.    |
| Clean CSV                  | Include the final 1,000-record CSV.                             |
| Power BI report            | Include the completed .pbix/dashboard.                          |
| Final PDF report           | This document; add final visuals/screenshots before submission. |
| Video presentation         | At least 5 minutes; add the video drive link inside the PDF.    |
| ZIP package                | Place required files in one folder and compress to ZIP.         |

**The project brief states that the final PDF should summarize the business problem, methodology, insights, model results and recommendations, and that the video link should be included in the PDF. Add the final dashboard screenshots and video link before packaging the submission.**

## Appendix A — Visuals to Add Before Submission

Use this page as a practical insertion checklist. The PDF has been intentionally prepared with visual placeholders so that the final Power BI screenshots can be inserted after you finish the report.

- Final Overview dashboard screenshot
- Price Distribution visual
- Rating Distribution visual
- Price vs Rating / K-Means scatter plot
- Model Analysis dashboard screenshot
- Key Influencers AI visual
- Word Cloud custom visual
- Books by Cluster
- Average Price by Cluster
- Average Rating by Cluster
- Book Details drill-through screenshot
- Bookmark / navigation screenshot
- Sync Slicer screenshot
- RLS View As screenshot

RECORDED VIDEO: <https://drive.google.com/file/d/1SpQEocTIVy3A-c3l6HdNP5wt0Yo83OMW/view?usp=sharing>

